



Matrices

Key terms

KSBs: K13, S11, B1, S53

matrix, equality, addition, scalar, multiplication, determinant, cofactor

9.1 Matrices

We have seen how numbers can be stored and manipulated in vectors. Matrices are an extension of this to larger grids of entries.

Definition 9.1 - Matrix

A **matrix** is a grid of entries which can be numbers, mathematical expressions or symbols.

Matrices are arranged in rows and columns and enclosed by round () or square [] brackets. For example:

This matrix contains 4 entries arranged in 2 rows and 2 columns.

The **dimension** or size of a matrix is written as $m \times n$ where m and n are the number of rows and columns of the matrix respectively. We might say it is an $m \times n$ matrix.

If $m = n$, so the number of rows and columns are equal, we say that the matrix is a square matrix.

$$\begin{array}{c} \leftarrow \text{columns} \rightarrow \\ \begin{array}{c} \uparrow \\ \text{rows} \\ \downarrow \end{array} \left(\begin{array}{cccc} \square & \square & \dots & \square \\ \square & \square & \dots & \square \\ \vdots & \vdots & \ddots & \vdots \\ \square & \square & \dots & \square \end{array} \right) \end{array}$$

Matrices are typically given names that are single letters, and the entries of a matrix are labelled with the letters a_{ij} .

An individual entry would be denoted a_{ij} where i is the row index reading from top to bottom and j is the column index reading from left to right.

Example 9.1. If A is an $m \times n$ matrix, then

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The entry in the

would be

9.2 Operations on Matrices

Definition 9.2 - Matrix Equality

We say that an $m \times n$ matrix A and an $p \times q$ matrix B are **equal** and write if and only if **both** of the following conditions are satisfied:

- they have the
- for all $1 \leq i \leq m$ and $1 \leq j \leq n$,

Definition 9.3 - Matrix Addition and Subtraction

Let A and B be two $m \times n$ matrices. The sum of two $m \times n$ matrices A and B is an $m \times n$ matrix $A + B$ defined by:

where $1 \leq i \leq m$ and $1 \leq j \leq n$.

Similarly, we can define the difference of two matrices using matrix subtraction:

where $1 \leq i \leq m$ and $1 \leq j \leq n$.

Example 9.2. The sum of two 2×2 matrices is given by matrix addition:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} =$$

Similarly, the difference is:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} =$$

Note

The sum and difference of two matrices of different sizes is

Exercise 9.1. Evaluate the following:

$$i) \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$ii) \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 7 \\ -11 \\ -13 \end{pmatrix}$$

$$iii) \begin{pmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \end{pmatrix} + \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

For all $m \times n$ matrices A, B and C , the following conditions are satisfied:

- $A + B =$
- $A + (B + C) =$

Definition 9.4 - Scalar multiplication

Let A be a matrix and k be a scalar. The product of the scalar and the matrix, kA , can be found using **scalar multiplication**, defined as:

i.e., we multiply each element in the matrix by the scalar.

$$k \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}.$$

Exercise 9.2. Evaluate the following:

$$(i) 2 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$(ii) \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 3 & 2 \\ 4 & -2 \end{pmatrix}$$

$$(iii) a \begin{pmatrix} 1 & 6 & 4 \\ 0 & 3 & -1 \end{pmatrix}$$

$$(iv) 101 \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} - 99 \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Let A and B be two $m \times n$ matrices, and k and ℓ be two scalars. Then

- $kA =$
- $k(A + B) =$
- $(k + \ell)A =$
- $k(\ell A) =$
- $-1 \times A =$

Definition 9.5 - Matrix Transpose

The **transpose** of an $m \times n$ matrix A is an $n \times m$ matrix denoted by A^T formed by switching the rows and columns of A , i.e.,

Transposing a matrix switches the rows and columns around so that row i becomes column i and column j becomes row j . i.e.,

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}^T = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

Example 9.3.

$$\begin{pmatrix} 2 & 0 & 3 \\ 4 & 1 & 5 \end{pmatrix}^T = \begin{pmatrix} & \\ & \\ & \end{pmatrix}.$$

Note that what was previously the _____ is now the _____ and so on. The number of rows and columns in the matrix have been _____

Exercise 9.3. Evaluate the following:

(i) $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}^T$

(ii) $\begin{pmatrix} 1 & 0 & -2 \\ 3 & -4 & 1 \end{pmatrix}^T$

(iii) $\begin{pmatrix} 2 & 3 & 5 \end{pmatrix}^T$

Let A and B be two square $n \times n$ matrices and k a scalar. Then

- $(A^T)^T =$
- $(A + B)^T =$
- $(kA)^T =$

9.3 Matrix Multiplication

We have seen that we can perform the arithmetic operations of addition and subtraction on matrices, and multiply a matrix by a scalar. Here we will look at

Definition 9.6 - Matrix Multiplication

Let A be an $m \times n$ matrix and B a $p \times q$ matrix, the product AB is defined as

The product AB is only defined if $n = p$, i.e., the number of columns of A is the same as the number of rows of B , and the resulting matrix has dimensions $m \times q$, i.e., as the first matrix and the second matrix, i.e., a

Example 9.4. The following pairs of matrices can be multiplied together:

These pairs cannot:

To check whether matrix multiplication is defined for a given pair of matrices, one easy check is to place the matrices next to each other, e.g. underneath the matrices, and if the two matrices have the same number of columns and rows respectively, the matrix multiplication can be performed.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} \quad \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix} \begin{pmatrix} g & h & i & j \\ k & l & m & n \end{pmatrix}$$

Furthermore, the dimensions of the product matrix is given by the number of rows of the first matrix and the number of columns of the second matrix - in the examples above, we'd get a 2×2 matrix and a 3×4 matrix.

9.3.1 Multiplying two matrices

The technique used to multiply two matrices together requires us to work across the rows of the first matrix (the i index) and down the columns of the second matrix (the j index).

Example 9.5. Consider the product of the two 2×2 matrices A and B given below:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

The first thing we need to do is _____ for these matrices.
They are _____ so this works.

Using the rule given in the matrix multiplication equation to calculate _____ (the $(1,1)$ th entry in the product AB), we use

We move through the elements from _____ and _____ multiplying the corresponding values and summing the result:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square \\ \square & \square \end{pmatrix}$$

We can write this as:

Carrying on across the top row of our product matrix AB : next we need to calculate the value of

Now our row number is _____ and column number is _____ so we multiply the values from _____ and _____ and sum the result.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & \square \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & \square \\ \square & \square \end{pmatrix}$$

Or in other words,

We can continue this process to fill in the rest of the matrix.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \square & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ \square & \square \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & \square \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & \square \end{pmatrix}.$$

Multiplying together larger matrices simply involves adding together the products of
and repeating this for each entry in the product matrix.

This is why we require the number of columns in the first matrix to be the same as the rows in the second: we
need to multiply together and add the products.

Exercise 9.4. *Given the matrices*

$$A = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 3 \\ 1 & 5 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 3 & -2 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

calculate the following (where possible):

(i) AB ;

(ii) BC ;

(iii) CD ;

(iv) CC^T

The following properties hold for matrix multiplication:

- $AB \neq$
- $A(BC) =$
- $A(B + C) =$
- $(A + B)C =$
- $(AB)^T =$

9.4 Determinants

A **determinant** is a that is calculated using the elements of a (non-square matrices do not have a determinant). Determinants play a very important role in linear algebra, and tell us about the properties of the matrix. One of their uses is they enable us to determine if a system of linear equations has a (as we will see later).

9.4.1 Calculating the determinant of a 2×2 matrix

Definition 9.7 - Determinant of a 2×2 matrix

The determinant of a square matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, denoted by $\det(A)$ or $|A|$, is

i.e., the product of the entries on the minus the product of the

Note

The formula for the determinant, will be a useful thing to remember. Come up with your own way of remembering it!

Example 9.6. The determinant of the matrix $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ is given by:

Exercise 9.5. Calculate the following determinants:

$$(i) \begin{vmatrix} 5 & 2 \\ 3 & 4 \end{vmatrix}$$

$$(ii) \det \begin{pmatrix} a & b \\ ka & kb \end{pmatrix}$$

$$(iii) \begin{vmatrix} 2 - \lambda & 3 \\ 5 & 6 - \lambda \end{vmatrix}$$

9.4.2 Determinant of $n \times n$ matrices

Computing the determinant of a 2×2 matrix is relatively straightforward, but computing the determinant of a matrix gets more complicated.

Essentially, we split the larger matrix up into so we can use the equation for a 2×2 determinant repeatedly to obtain the overall determinant. This is done using **minors** and **cofactors**.

Definition 9.8 - Minor

The **minor** of an element of an $n \times n$ square matrix is denoted by and is the determinant of the that is formed by

Example 9.7. Given the matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

then the minor M_{21} is the determinant of the matrix A with

$$M_{21} = \begin{vmatrix} \square & a_{12} & a_{13} \\ \square & \square & \square \\ \square & a_{32} & a_{33} \end{vmatrix} =$$

Exercise 9.6. Given the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

calculate:

(i) M_{11} ;

(ii) M_{12} ;

(iii) M_{13}

9.4.3 Cofactors

Definition 9.9 - Cofactor

The **cofactor** of an element of a square matrix is denoted by C_{ij} and is defined as

The $(-1)^{i+j}$ term in this equation is C_{ij} and
which results in the following pattern of signs:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Example 9.8. Given the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Then

Now we have the terminology of minors and cofactors, we can use them to calculate determinants. For a larger matrix, we choose a single row or column of the matrix, and work our way along it.

Definition 9.10 - Determinant of an $n \times n$ matrix

The determinant of an $n \times n$ matrix A is defined by

for some fixed value k in the range $1 \leq k \leq n$ which represents a

This equation allows us to express the determinant of an $n \times n$ matrix in terms of

We can then apply the formula again to the sub-matrices. Continuing in this fashion we will eventually just be calculating determinants of 2×2 matrices, which we know how to do.

Example 9.9. To calculate the determinant of the matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

we can

- this means we are using

This

is just the of the determinants of each multiplied by the entry defining which rows and columns are not used in that matrix.

We could also have chosen to expand along and the

In our first example, the final line can be multiplied out to give $aei - afh - bdi + bfg + cdh - ceg$, and this is the same (after some rearrangement) as the result when we expand along column 2.

Note

row or column we expand along to compute the determinant - we will always get the same answer. It is usually preferable to expand along the row or column with the most zeros or the row or column with the most non-zero elements to simplify the calculations.

For larger matrices we have to apply the equation recursively until we get to 2×2 determinants where we can use our 2×2 determinant equation.

Exercise 9.7. (a) Calculate the determinant of the matrix

$$\begin{pmatrix} 1 & 0 & 4 \\ 2 & 5 & 6 \\ 4 & 5 & 2 \end{pmatrix},$$

by expanding along:

(i) row 1;

(ii) column 3

(b) Calculate the determinant of the 4×4 matrix

$$\begin{pmatrix} 1 & -1 & 4 & 3 \\ 2 & 0 & 5 & -3 \\ 1 & 2 & 4 & 5 \\ 2 & 0 & -2 & 4 \end{pmatrix}$$

Determinants have the following properties:

- $\det(AB) = \det(A)\det(B)$
- $\det(A) = \det(A^T)$
- If a matrix M has a row or column with all zero elements then $\det(M) = 0$
- Interchanging any two rows of a matrix changes the sign of the determinant